

initial-state space
Sobol-sampled initial states

**human corrections — steer back early, before failure,
or rescue the policy after it has left the funnel**

every trajectory — including failures —
trains the **critic Q**; only curated human
data trains the **actor π**

precision bottleneck
(e.g. the insertion)

**task
success**

**at the bottleneck, most sampled action chunks miss —
the critic Q executes the one that threads it (best-of-N)**

autonomous failure — catastrophically off-course
(kept as negative data for the critic)

**counterfactual demo — a fresh human demo from the same failed initial state,
showing what the policy should have done instead**

— policy rollout — human correction — counterfactual demo ✗ failure
----- sampled action chunks (N) — chunk chosen by the critic ■ precision bottleneck

